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ABSTRACT

Beetroots are source of nitrate and phytochemical compound that take ascorbic acid, carotenoids, phenol acid, and flavonoids and Yogurt is a fermented dairy product that contains live cultures of lactic acid bacteria, which contribute to its probiotic properties that lead us to make project. This term paper presents a comprehensive study investigating the effects of red beetroots juice extract on phenol content and sensory characteristic of Yogurt. The study is structured into five specific objectives: (1) to make beetroot juice and application beetroots juice in yogurt production (2) Determination the presence of Total polyphenol content in yogurt added with beetroot juice and (3) Sensory evaluation. Each objective is meticulously designed and executed to provide a thorough understanding of how the addition of beetroot juice influences the yogurt product sensory attributes and affect physicochemical properties of yoghurt samples where we had two sections non sweetened yoghurt which was section A and sweetened yoghurt which was sweetened yoghurt samples obtained results showed that there was significant different among sample of section A where p Value obtained was 0.005477 which was below acceptable p value for significant

Target (B)

DEPARTMENT: AGRICULTURAL ENGINEERING
OPTION: FOOD PROCESSING
ACADEMIC STUDY: 2022-2023

IMPROVING MINERALS OF TOMATO KETCHUP USING RED BEETROOT'S POWDER

PREPARED BY:
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REG N°: 20RP09099

Supervised by: Dr. MUKUNZI Daniel

THE RESEARCH PROJECT IS SUBMITTED TO THE DEPARTMENT OF AGRICULTURAL ENGINEERING IN PARTIAL FULFILLMENT OF THE AWARD OF THE ADVANCED DIPLOMA CERTIFICATE IN FOOD PROCESSING OF RWANDA POLYTECHNIC

Musanze, October 2023

ABSTRACT

This project had aim of improving nutritional profile and improving sensory characteristics of tomato

different and for section B samples the results of sensory evaluation showed that there was significant different where obtained value was p value of 0.003429 which in the range acceptable p value < 0.05 of significant different and for nutritional analysis results showed that samples for section A varied in the viscosity where S0 had 0.925, S1 had 1.199, S2 had 0.973 S3 had 1.061 and S4 had 1.077 of viscosity and for section A samples variation in the viscosity was S0 had 1.9955, S1 had 2.106, S2 had 1.718, S3 had 1.304 and S4 had 1. 6525 of viscosity.

DECLARATION

We, MANIRAFASHA Patrick, NIYONKURU Clementine here with declare that, this research project entitled 'Effects of Red beetroot juice extract on phenol content and Sensory characteristic of Yogurt' is our original work and has not been previously submitted for award of a degree in any other university or institution of higher learning and any work quoted herein, is indicated and acknowledged

ketchup using beetroot powder where all activities carried out to reach on the objective of the study were preparation beetroot powder where beetroots bought from market of Musanze and sent into food processing workshop at RP-IPRC Musanze for production of beetroot powder and also tomatoes bought from Musanze market and prepared well to make pulp to be used for ketchup manufacturing. In making tomato ketchup with beetroot powder there were different rations of different ingredients used like vinegar, honey, salt , garlic and also onions in addition beetroot powder where obtained samples analyzed and used in sensory evaluation to ensure their level of acceptability, according to the obtained results in sensory evaluation there was significant different in samples which was P of 0.010829 and it was less than alpha 0.05 that meant that there was significant different among samples. Furthermore the laboratory analysis results showed that the addition of beetroot powder in tomato ketchup making affected the moisture content level where decreased as the amount of red beetroot powder increased where sample 0 had 81%, sample 1 had 75% and sample 2 had moisture content of 69%. Addition of red beetroot powder in ketchup samples increased ash content where sample 0 had 2.7% but it did not contain beetroot powder and sample 1 had 4.9% ash content where sample 2 had 6 % of ash content.

Study Hypothesis: Adding beetroot juice to yogurt enhances its sensory appeal and acceptability.

by means of a list of reference.

Students Names

Name: NIYONKURU Clementine

Signature: ……………………………… Date ……………………..

Name: MANIRAFASHA Patrick

Signature………………………… Date ………………………

This report has been written and submitted after approval of supervisor Dr, Daniel MUKUNZI Lecturer of food processing program in agricultural engineering in RP-IPRC MUSANZE

Signature… …………………………Date ……………………

Mr. NGENDO Clement

Head of department of Agriculture engineering in RP-IPRC MUSANZE

Signature ……………….Date…………………………

DECLARATION

This is to certify that this memory research project work titled, "improving minerals of tomato ketchup using red beetroot's powder" have been carried out by CHANCE Pamela and Diane IRAKOZE is our own original work and has not been presented or submitted in any other university or any higher learning institution for any degree award.

Names of students:

CHANCE Pamela

Signature…………………………………

Date…………………………………

IRAKOZE Diane

Signature…………………………………

Date…………………………………

CERTIFICATION

This is to certify that the work untitled "improving minerals of tomato ketchup using red beetroot's powder" was carried out by IRAKOZE Diane 20RP09099, and CHANCE Pamela 20RP05365 .

Supervisor:

Dr. MUKUNZI Daniel

Signature…………………………………

Date……………………………

Head of department:

NGENDO clement

Signature………… ……Date……………………………

DEDICATION

DEDICATION

This project is gratefully dedicated to:

To Almighty God.

To our parents.

To our brothers and sisters.

To all member RP/ IPPC – MUSANZE.

To our relatives.

To our friends.

To our entire classmate.

To all individuals whose assistance has been useful during our studies.

To our lecturers.

ACKNOWLEGEMENT

First and foremost, we address sincere heartfelt gratitude to Almighty God for his having kept us alive before and during our research project activity and through the whole period our studies. We are very grateful to whole RP/IPRC MUSANZE academic staff, particularly Agriculture engineering department especially food processing lectures for providing professional support and their

This project is gratefully dedicated to:

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To All Member of RP/ IPPC – MUSANZE;

To Our Relatives;

To Our Friends;

To All Our Classmate;

To All Individuals who's Assistance Has Been Useful during Our Studies

ACKNOWLEDGEMENTS

First and foremost, we thank the almighty god for his abundant blessings and protection during all our studies. We are also thankful to RP/IPRC MUSANZE authorities for providing all necessary facilities during the research work. We acknowledge all the academic staff, especially, the lecturers of agriculture engineering department for their kind assistance in both practical and theoretical knowledge provided.

Much gratitude goes out to our supervisor Dr. MUKUNZI Daniel for his efforts and energy in guidance, collaboration, constructive suggestions, encouragement and his dedication, which helped us to come to the successful completion and advice for this work.

We are grateful to our classmate and all well-wishes who helped us to conduct our activities, we heartfelt thanks go to our dear parents, colleagues and friends who have shown great understanding and sympathy since we have been involved in the preparation of this research.

We also extend our deep appreciation to the government of Rwanda and Rwanda polytechnic, guidance at various stages during our studies. We are very grateful to MASTER CURD FOUNDATION to their support to us during our research project activities.

Much gratitude goes out to our supervisor Mr. MUKUNZI Daniel for his efforts and energy in guidance, collaboration, constructive suggestions, encouragement and her dedication, which helped us to come to the successful completion and advice for this work.

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We also extend our deep appreciation to the government of Rwanda and RWANDA POLYTECHNIC, IPRC MUSANZE for their facilities they provided to accomplish this research project. It is with deep respect and gratitude that we acknowledge everyone whose contribution has brought us this far.

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1. Yogurt
2. Beetroots
3. Milk
4. Phenolic LIST OF ABLEVIATION

Dr: Doctor

Mr.Mister RP: Rwanda Polytechnic IPRC: Integrated Polytechnic Regional College pH :Potential Hydrogen

0c : Degree Celicius

ML: mille litre

%: Parentage

OEL: Occupational exposure limit

P- value :probability value

S: samples

CPS : Centipoise

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ABBREVIATION

RP: Rwanda polytechnic.

IPRC: Integrated Polytechnic Regional College.

Reg n o: Registration number.

Mr.: Mister. 0c: Degree Celsius g: Gram

%: Percentage

S1: Sample1

S2:Sample2

HACCP: Hazard Analysis Critical Control Point

LIST OF FIGURE

Vit: Vitamins

Ts: Total sample

As: average sample

Keywords: Beetroot, Ash content, Nutritional Enhancer, Health benefits, Product development, Tomato, Ketchup

CHAPTER ONE: GENERAL INTRODUCTION

1.1. Background

Ketchup has a long and interesting history, with origins that can be traced to ancient china. Over the years, ketchup has evolved and changed to become delicious and versatile condiment that we know and love today (woods, 2016).Ketchup is food product made from tomatoes which may help to fight against cancer, protect human heart and brain and as well as offer fertility support to men(Snyder, 2022). Ketchup is a type of sauce that is typically made from tomatoes, vinegar, sugar and variety of spices and seasoning. In food processing, ketchup is typically made by cooking down tomatoes, adding other ingredients, and then blending them together to create smooth, thick sauce.

Red Beetroots are vegetables are high in different nutrients like fiber, vitamin B9, and vitamin, minerals like potassium, iron, and manganese. Thus leading to offer different health benefits on human health,

CHAPTER ONE. INTRODUCTION

1.1. Background of the study

Yogurt defined as "the coagulated milk product obtained by lactic acid fermentation through the action of Lactobacillus delbrueckii ssp. bulgaricus (Lb. bulgaricus) and Streptococcus thermophilus from milk and milk prod. attractive Yogurt has highly nutritional properties; it is low in calories (around 90 kcal per serving) but contains enough macro- and micronutrients (proteins, fatty acids, calcium, phosphorus, and vitamins) to cover a person's daily needs Typically, yogurts are produced at temperatures around 42°C, which promotes the optimal growth of both S. thermophilus and Lb. bulgaricus (Nagako, 2019). When milk is inoculated with these two bacteria, they usually grow in succession and S. thermophiles presents diauxic growth, the S. thermophilus first grows exponentially (for the first 90 to 120 min) and then experiences a short latency period. Lb. bulgaricus stays at inoculation levels. S. thermophilus subsequently resumes growth, albeit at a reduced rate, and Lb.

like to reduce blood pressure, increase the exercise capacity (Bjarnadottir, 2023). Red beetroots is source of Natural sugar and color, fiber, folate (vitaminB9), manganese, potassium , iron and vitamin, but ketchup is lack in natural sugar and some nutritional profile therefore, mixing ketchup with red beetroot, will produce nutritional balanced food.

One promising approach to enhancing nutritional profile of tomato ketchup is through the incorporation of red beetroot. Red beetroot is root vegetable known for its vibrant color, earthy flavor, and high nutrient density. It is rich in essential vitamins, including vitamin C, vitamin B6, and folate, as well as minerals such as iron, potassium, and magnesium. Furthermore, red beetroot contains dietary fiber, which is crucial for digestive health, natural and overall well-being(Jeong, Lee, & Park, 2023). The objective of this study is to investigate the potential of red beetroot as a natural ingredient for improving the nutritional profile of tomato ketchup. By incorporating red beetroot into tomato ketchup formulation, we aim to enhance its nutrient content while maintaining its sensory characteristics and consumer acceptance. This research aims to explore the feasibility and benefits of enriching tomato ketchup with red beetroot, thereby providing a healthier alternative to consumers without compromising the familiar taste and appeal of this beloved condiment.To achieve this objective, evaluated the nutritional composition of the red beet rootenriched tomato ketchup, where ash content obtained was increased as the amount of beetroot powder increased and Sensory evaluation was bulgaricus starts to grow exponentially. As the milk becomes acidified, reaching a pH of around 5.2, S. thermophilus stops growing. In contrast, the growth of Lb. bulgaricus continues until pH levels drop to around 4.4 (Smith, 2015)

Yogurt should contain a minimum of 2.7% (m/m) milk proteins, a maximum of 15% milk fat, a minimum of 0.6% titratable acidity (expressed as % of lactic acid), and a minimum of 107 CFU/g of microorganisms (total microorganisms in the starter culture). Yogurt has highly attractive nutritional properties; it is low in calories (around 90 kcal per serving) but contains enough macro and micronutrients (proteins, fatty acids, calcium, phosphorus, and vitamins) to cover a person's daily needs

In addition to its unique flavor comes from to its acidity, the yoghurt is flavored with extracted fruit flavors from strawberry, vanilla, pineapple etc, and can also be flavored by vegetables juice or flavors.

In these above contexts, the projects will be focused on beetroot vegetables juice addition to the yoghurt in order to combine benefits of beetroot vegetables and yogurt. Beetroots contain a wide variety of essential nutrients for a healthy life; they are rich in fiber, folate (vitamin B9), manganese, potassium, iron, and vitamin C and also contain nitrates, which can help lower blood pressure and improve exercise performance(Jeong, Lee, & Park, 2023).. Beetroots are known for their earthy flavor, surprising sweetness and the taste of

conducted to assess the acceptability and preference of the enriched ketchup compared to the traditional version that research resulted to the significant different among samples where obtained p value was 0.010829. Through these analysis, determined that beetroot can successfully improve the nutritional value of tomato ketchup, offering a healthier option for individual seeking to make more nutritious dietary choices.Overall, this study holds significant potential for the food industry, consumers and public health. By enhancing the nutritional profile of tomato ketchup using red beetroot, we can contribute to promoting healthier eating natural ingredients habits, diversify food options, supporting local farmers and suppliers, and foster food innovation. Therefore incorporating ketchup with red beetroot as natural enhancer it will provide natural sugar content and also improve nutritional profile and reduce chemical enhancers to the consumers.

1.2. Problem statement

Ketchup is most commonly used adjunct for snacks, in house hold and also in different restaurant. Nevertheless, since tomato ketchup is commonly used ketchup the nutritional value and apparent bio functional properties of the ketchup are limited to the nutritents present in the tomato and its stability after processing. In addition ketchup is one of tomato based product mostly liked in the world with ash content varied where ranged from 1.81 to 5 % of when compared to the commercial samples of 2.95% ash content (Prakash, 2016). Furthermore beetroot are roots which are rich in different minerals like manganese which is good for bone health, magnesium, potassium, sodium, beets can also vary depending on their variety for example, golden beets are sweeter than red beets(Bjarnadottir, 2023).

Key words: yogurt, polyphenol, beetroots and milk.

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1.2. Problem statements

Yogurt is the product which is contain a lot of nutrients in generally it was not contain some amount of polyphenol which can help the peoples who consume that products to prevent some dangerous diseases like Cancer, Heart diseases and Oxidative Damage. So Red beetroots juice contains a specific type of polyphenol called Betaines which contribute to its health benefits such as heart health, antioxidant properties, and Liver health and antiInflammatory in addition that polyphenol contains a chemical compound of pigments that can help yogurt to be colored and flavored by red beetroots juice. In addition red beetroot juice used us natural coloring and flavoring agent in yogurt product.

1.3. Objectives of the study

1.3.1. General Objective

To study the effects of beetroot juice on phenolic content and sensory characteristic of yogurt

1.3.2. Specific objectives

- To process beetroot juice and application of processed beetroots juice in

phosphorus, iron, zinc, copper, boron, silica and selenium (Nistor & Ceclu, 2020). Incorporation of beetroot powder in ketchup will enhance the mineral values in the produced product which may be higher than commercial ketchup

1.3. Objective of the study

1.3.1. General objective

To improving minerals of tomato ketchup using red beetroot's powder

1.3.2. Specific objectives

- To produce red beetroot powder
- To produce tomato ketchup incorporated with red beetroot powder
- To determine ashl content of produced ketchup
- To determine sensory characteristic of processed red beetroot tomato ketchup

1.3. Hypothesis

Incorporating red beetroot with tomato ketchup improves nutritional profile by increasing the content of minerals considered as ash content while maintained the desired sensory characteristics and natural ingredients to meet with consumer acceptance.

1.4. Significance of the study

This study is very important because it deals with improving the nutritional benefit of tomato ketchup incorporating with red beetroot will help in improving nutritional content where the amount of ash content, in addition will help cultivators to obtain market of yogurt processing

- To determine the Presence of total polyphenol in yogurt added with beet root juice
- To determine the sensory evaluation of yogurt added with beet root juice

1.4. Hypothesis of the study

- Yogurt added with beetroots juice improved acceptable sensory characteristics.
- Addition of beetroots juice in yoghurt was not increase of polyphenol but affected other physicochemical properties in yoghurt.

1.5 Significance of the Study

This research of effect of beetroots juice on phenolic content of yogurt will leads to a new alternative of the use of chemical flavoring and coloring agents in yogurt which have negative effects on human health as well as by introducing a new way of combining health benefits of yogurt and reduction of beetroots vegetables losses. The project of using beetroots juice in yogurt making also led to the production of yogurt which has distinctive sensory characteristics and nutritional value from the existing ones. In addition will improve quality of yogurt because that yogurt will contain amounts polyphenol which has positive effect on human body.

1.6. Scope

The research aims is to investigate how the addition of beetroot juice affects the phenol content of yogurt.The study will explore variations in phenolic content. by using different their beetroots as well as tomatoes, this project will help to bring the product with higher minerals content than commercial ketchup.

1.6. Scope of the study

This study took place at RP-IPRC Musanze in food processing workshop where the raw materials used were bought in Musanze market. This research also focused on the nutritional content of produced ketchup which are ash content, folate , fiber and moisture content were determined as well as sensory evaluation done by using 9 trained panelists and sensory evaluation method used was 9 point hedonic scale, this research conducted for 6 months.

CHAPTER TWO: LITERATURE REVIEW

2.1. Tomatoes

Tomatoes are fruit that belong to the nightshade family. They are typically red and round, although they can come in other colors and shapes, such as yellow or pear shaped. Tomatoes have a slightly sweet and tangy flavor and are often used in cooking especial in sauces, soups, and salads. They are also good source of vitamin C, potassium, and other nutrients(Bjarnadottir Adda, 2023). Furthermore Tomatoes are rich in different nutrients like vitamin and minerals which offer different health benefits(Gaur, 2023)

2.2.1. Composition of tomato

Basic composition of tomato is as follow:

Table 1Composition of tomato

concentrations or ratios of beetroot juice in yogurt formulations. in addition the research will assess the impact of incorporating beetroot juice into yogurt on its nutritional composition. This will involves in analyzing the viscosity and change in Polyphenol

This research will take place in Musanze district, Nkotsi sector, in IPRC Musanze food processing work shop. The raw materials like milk will bought from a farmers in Nkots sector and other raw materials will bought from Musanze marke ts as well as in Kigali super markets. the research will done from April up to October 2023

CHAPTER TWO: LITERATURE REVIEW

2.1 Yogurt

Yoghurt is known as fermented daily product obtained by lactic acid fermentation where milk sugar called lactose converted into lactic acid by the action lactic acid bacteria LAB which are streptococcus thermophiles and lactobacillus delbrueckii subsp. Bulgaricus (Nagako, 2019).

2.1.1. Yoghurt Nutritional content

Table 1: Yoghurt Nutritional content

Nutrients	Amount per 100 grams
Calories	97

Nutrients	Amount
Water	95%
carbohydrates	3-4 %
Fat	1 %
proteins	1.2%
Fiber	1%
Vitamin C	20 %
Potassium	10%

2.2.2. Benefits of tomatoes

Antioxidant content: Tomatoes are rich in antioxidants, including lycopene, which may help protect against certain types of cancer and cardiovascular disease, Vitamin C: tomatoes are good source of vitamin c, which is important for immune function and skin health, Low in calories: tomatoes are low in calories and can be used in a variety of dishes, making them a versatile ingredient for a healthy diet, Potassium: tomatoes are also good source of potassium, which is important for maintaining health blood pressure and heart function(Fornelli, 2007).

Versatile ingredient: tomatoes can be used in a variety of dishes, including sauces, soups and salads, making them versatile ingredient that can be incorporated into health diet in many different

Moisture 77.2 % Protein 3.9% Total fat 3.5% Ash 0.2 % Carbohydrate 15.5 % Total sugar 4 Vitamin B12 32.25% Phosphorus 19.29% Selenium 17.64% Calcium 10% Zinc 6.62% Carbohydrates 3.66% Potassium 3% Thiamine 1.92% Copper 1.89% Niacin 1.30% Folate 1.25%	ways(Snyder, 2022). May help to prevent type 2 diabetes: The function of lycopene in human body is to contribute to the avoidance of type 2 diabetes because it has ability to protect cells from damage and also may help in minimizing inflammatory, and also increase the body defense mechanism(Michelson, 2023) May reduce cancer risks: Tomatoes are rich in lycopene and beta-carotene those two items called antioxidants present in tomatoes may help to act as anticancer due to the anticancer properties containing(Blum, Monir, & wirsansky, 2005) May support male Fertility: Tomatoes are rich in lycopene when consumed the level of lycopene in male increases thus leading to increase of blood lycopene, the movement of sperm, and it is indicator of fertility(Cruz Bojorrquez, Gonzlez, & Sanchez, 2013). 2.2. Tomato ketchup Tomato ketchup is popular condiment enjoyed by millions of people worldwide. Its tangy and sweet flavor makes it a versatile accompaniment to various dishes, ranging from burgers and fries to sandwiches and hot dog. However, despite its widespread consumption, traditional tomato ketchup often lacks significant nutritional value, primarily consisting of sugar, salt and tomato concentrate.in addition Ketchup is a condiment typically made from tomatoes, vinegar, sugar and spices. It is often used as a topping for burgers, hot dogs, French fries and other foods. The origin of ketchup can be traced back to china, where a similar sauce was made from fermented fish. The sauce was letter brought to Europe by traders and eventually made its way to

This table is showing yoghurt nutrition content (Kumar, 2020).

2.1.2. Yoghurt Health benefits

Yoghurt is nutritious product reason why it has various health benefits on human health and human body, where help in prevention, assist in maintaining body health.

Promote digestive health: yoghurt help in restoring balance in intestinal flora which help in your body to treat antibiotics associated diarrhea by the action of active and live cultures present in yoghurt, not only that function because probiotics help to minimize symptoms of lactose intolerance by helping in digestion of lactose.

Help people with problem of lactose intolerance: the problem of lactose intolerance is caused by the less ability of human body to digest lactose. In yoghurt manufacturing the level of lactose intolerance is lowered, by converting lactose into glucose and galactose, where glucose is converted into lactic acid by the action of lactic acid bacteria

Lower risks of abnormal blood pressure: abnormal blood pressure it could be high blood pressure this may be the fact to cause problem of heart diseases, taking yoghurt may be wise thing to help you to fight against heart diseases.

Improve the bone health: yoghurt contain

America, where it was adapted to include tomatoes. Today, ketchup is popular condiment around the world, with many different variations and flavor profiles.

In recent years, there has been a growing emphasis on promoting healthier eating habits emphasis on promoting healthier eating habits and incorporating nutrients rich foods into our diets. As a result, there is pressing need to improve the nutritional profile of commonly consumed food products, including condiments like tomato ketchup. By enhancing the nutritional content like vitamin and minerals, we can contribute to improving public health and supporting individuals in making healthier food choices (Elise M, 2013).

2.2.1. Types of tomato ketchup

They are many different of tomato ketchup available, each with its own unique flavor and texture. Some common types of tomato ketchup include:

Regular ketchup: this is most common type of ketchup, made from tomatoes, vinegar, sugar, and spices.

Organic ketchup: this type of ketchup is made from organic ingredients and does not contain any artificial preservatives or additives.

Low-sugar ketchup: this type of ketchup is made with less sugar than traditional ketchup making it healthier option for those who are watching their sugar intake.

Spicy ketchup: this type of ketchup is made with additional spices or hot peppers, giving it a spicy kick.

nutrients which are calcium and proteins which are responsible for preventing diseases of osteoporosis which is shown by weak bones, and brittle bones mainly in elder people or adults and also this problem increase as they get more age and that nutrients in yoghurt help in making higher bones density

Yogurt is a fermented dairy product that contains live cultures of lactic acid bacteria, which contribute to its probiotic properties. It is also a significant source of high-quality proteins , vitamins(e.g. ., riboflavin,B12), minerals (e.g. calcium, phosphorus), and other essential nutrients regular yogurt consumption has been associated with improved digestion, enhanced immune function, and reduced risk of chronic (Arnarson, 2019).

2.2 Beetroot

Beetroot is most important vegetables grown sub-tropical and tropic area of the world. Beetroots are source of nitrate and phytochemical compound that take ascorbic acid, carotenoids, phenolic acid, and flavonoids (kavitha & Smetanska, 2020).

2.2.1 Composition of beetroot juice

Beetroot juice is known for its high nutritional value, offering several health benefits. It is a rich source of dietary fiber, vitamins, (e.g., vitamin C, folate), minerals (e.g., potassium, iron), and antioxidants. Studies have reported its potentiall role in reducing blood pressure, improving exercise performance, enhancing cognitive function, and supporting cardiovascular health row red beetroot contain

Fruit ketchup/Vegetable ketchup: some ketchup are made with fruits or vegetables like apples or peaches and beetroot which can give them a unique flavor and texture.

Gourmet ketchup: this type of ketchup is often made with high-quality ingredients and may feature unique flavor combinations like truffle or bourbon.

2.2.2. Health benefits of Tomato ketchup

High lycopene content: lycopene is a powerful antioxidant that can help you to protect against certain types of cancer and heart disease. Tomato ketchup is convenient way to consume more lycopene in your diet.

Good source of Vitamins A and C: Tomatoes are good source of vitamins such as A and C, Which are important for maintaining health skin, eye, and immune function. These vitamins also act as antioxidants, helping to protect against cellular damage and inflammation.

Low in calories: Tomato ketchup is low in calories, making it a good condiment choice for those who are watching their weight.

Versatile: it can be used variety of dishes to add flavor and nutrition.

Low in fat content

2.3. Beet root

Red beetroots are known as root with dark-red purple color that can be eaten either raw like salad or further processed. Red beet may be having distinctive color due to the presence of nitrogen-that containing water-soluble pigments betalains. They are mostly used as a natural food colorant and have E number of E-162. Red beetroot are rich in vitamins like vitamin C, vitamin B6, folate and more nutrient per 100g (T, Howatson, & Stevenson, 2015)

2.2.2. BEETROOT JUICE NUTRITION

Table 2: Nutrients in Beetroot juice

NUTRIENTS	AMOUNTS Per 100g
Water	91.3
Betalaine (ug)	127.7
Protein	1.89
Carbohydrate	7.23
Fiber	3.25
Sugar	6.76
Lipid	0.15
Ash	1.08
A-Carotene (ug)	22.0
B calotene(ug)	0
Lycopine	30
Luteine	0
Foliate	109

minerals dietary and also possesses has high antioxidants capacity(kavitha & Smetanska, 2020).

2.3. 1. Scientific classification of Red beetroot

Kingdom: Plantae
Clade: Angiosperm (flowering plant)
Clade: Eudicotis
Clade: Core eudicotis
Order: Caryophyllales
Family: Amarthaceae
Genus: Beta
Species: Vulgaris

The red beetroot belong to the Amarathaceae family, which includes many other plants such as spinach and quinoa. The genus for red beetroot is Beta, and the species is vulgaris.

2.3.2. Nutritional content of Red beetroot

Red beetroot is rich in nutritional profile. It packed with a variety of essential vitamins, minerals, and dietary fiber. Here is an overview of the nutritional content of beetroot per 100 grams:

Macronutrients of Red beetroot

Carbohydrates: 10g consists of main sugars (fructose and glucose) that provide energy.

Dietary fiber: 2 g per 100g. Which support digestive health, maintain healthy weight and regulate blood levels

Protein: red beetroot it is not particularly high in protein but it can provide small amount. (Bjarnadottir Adda, 2023).

Micronutrients of Red beetroot

Beetroot is rich in different nutrients as shown in the table 1 below

Niacin(mg)	0.334
VitamineB6	0.067
Sodium(mg)	78.0
Potassium(mg)	325
Phosphorus(mg)	40.0
Magnesium(mg)	23.0
Calisium (mg)	16
manganese(mg)	0.359
Zinc (mg)	0.365

This tables is showing the nutrition profile of red Beetroot (Ceclu & Nistor, 2020)

2.2.3. PRODUCTION OF BEETROOT JUICE

Reception of beetroot fruits with balance, table, bucket

Washing the fruits with clean water and bucket

peeling and blending with knives and blender

dilution of juice with clean water cups

cooking of juice with saucepan,oven,stirer

cooling with cooling container

Table 2: Nutrients in beetroot

Nutrients	amount
Vitamin C	4.9 mg
Folate	0.0109 mg
potassium	325 mg
magnesium	32 mg

2.3.3 Health and Benefit of Red beetroot

Red beetroot offers several health benefits due to its rich in nutritional profile. Here are some of potential health benefits associated with consuming red beetroot:

Antioxidant properties: A red beetroot contains various antioxidants, including betalains and vitamin C, which help protect cells from oxidative stress and reduce the risk of chronic diseases.

Health heart: the nitrates present in red beetroot are converted into nitric oxide in the vessels, promoting health blood flow and help lower blood pressure and reduce the risk of heart diseases.

Improved exercise performance: the nitrates in red beetroots have been found to enhance exercise performance.

Enhanced digestion: red beetroot is good for source of dietary fiber, which promotes health digestion and helps prevent by nourishing beneficial gut bacteria(williams, 2023)

Anti-inflammatory properties: the betalains in red

filitration with filter or muslum cloths
storage with storage tank or container

Figure 1: production of beetroot juice

2.3 General Yoghurt Processing Steps

• Adjust milk composition and blend ingredient: Milk composition were adjusted to achieve the desired fat content and total solids content. Often dry milk is added to increase the amount of whey protein to provide a desirable texture. Ingredient such as stabilizer.

• Pasteurize Milk: The milk mixture is pasteurized at 850c for 30 minutes or at 950c for 10 minutes a high heat treatment is used to denature the whey protein. This allows the protein to form more stable gel, which prevents separation of water during storage. The higher heat also minimizes the number of spoilage organism, in the milk to provide better condition for starter culture growth. The yoghurt is pasteurized before starter culture are added to ensure that the cultures remain active in the yoghurt after fermentation to act as probiotics (Melese, 2015)

• Homogenization: The homogenized (2000 to 2500ppm) to mix all ingredient thoroughly and improve yoghurt homogeneous consistence.

• Cool Milk: The milk is cooled to 420c to bring yoghurt to the temperature allow growth of the starter culture.

• Inoculate with Starter Culture: The mixed into cooled milk with 420c-450c

• Hold: The milk is held at 420c until

beetroot possess anti-inflammatory properties, which may help reduce inflammation is associated with various diseases, including diabetes, heart diseases and certain types of cancer(Hawkins, 2023)

Detoxification Support: the betalain pigments in red beetroot act as potent detoxifiers, helping neutralize and eliminate toxins from the body. This support liver

Brain health: Red beetroots contain natural nitrates that have been linked to improved cognitive function and blood flow to the brain. Regular consumption may help enhance mental performance and reduce the risk of age-related cognitive decline(Chen & Jin, 2021)

Keep blood pressure in check: beetroot facilitate in minimizing elevated blood pressure levels which may risk to heart diseases and also help to reduce the level of systolic and diastolic blood pressure(Coyle, 2023)

2.4. Beetroot powder

Beetroot powder is kind of fine powder obtained from grinding dried beetroot pieces which is nutritious due to be having all nutrients present in the fresh beetroot like betanin is one of color pigments which help to color beetroot with purplish red color, it sometimes used to enhance or increase the red color in the tomato paste, sauces, desserts, jams, jelly, ice cream, sweets and breakfast cereals. The reason behind beetroot powder production may be caused by that fresh vegetables and fruits cannot be stored for long time due to the higher moisture content reason why drying process help in making longer shelf life to those fruits or vegetables where pH4.5 is reached. This allows fermentation to progress to form a soft gel and the characteristics of yoghurt properties.

- cool: The yoghurt is cooled to 4-7oc to stop fermentation process
- Add Fruit and Flavor: Fruit and flavor are added to different steps depending on the type of yoghurt. For set yoghurt the is fermented I the cup. For Swiss style yoghurt the fruit is blended with the fermented, cooled yoghurt prior to packaging
- Package: The yoghurt is pumped from the fermentation vat and packaged as desired

2.3.1 Yoghurt stabilization

Stabilizer are commonly used in cultured products to control texture and reduce whey separation since they impart good resistance to syneresis and smooth sensation in the mouth feel by binding water to reduce water flow in the food matrix space (Amatayakul & Tamine, 2006). Two of the most frequently used stabilizer are gelatin and starch, the use of modified starch help to create a creamier texture but it should base on processing condition(shear, heat, pressure and pH) that will be encountered, usage level of in culture product are usually 0.1-0.4% for gums from 0.8% and above for modified starch and up 2% for starch(Melese, 2015).

Starch is also used in yoghurt to increase its viscosity, improve it mouth feel, and prevent syneresis, starch granules absorb water and well to many times their original size, resulting in increased viscosity of the solution(Verbeken, beetroots meet with drying process(Ling Ng & Sulaiman, 2018)

2.4.1. Factors affecting beetroot powder nutritional content

During beetroot powder production there some nutrients or pigments undergo lost due to different factors here are some factors may interfere in reduction of beetroot powder nutrients

Drying: drying process done at high temperature may affect some nutrients found in the fresh beetroot like color pigments where the higher temperature denature the color pigments where the resulted product should be obtained with different color due to higher temperature. Mainly drying should be done at temperature of 500 C for like 6 hours(Wlodarczyk & Janiszewska, 2013)

Drying process may be done spray drying, foam mat drying all of that has effect on the nutrients of beetroot.

2.5. Health benefits of beetroot powder

Beetroot powder is a rich source of diverse minerals such as potassium, sodium, phosphorous, calcium, magnesium, copper, iron, zinc and manganese It is commonly consumed powder, bread, gel, boiled, oven-dried, pickled, pureed or jam-processed across different food cultures(Whittaker, 2023)

Beetroot powders are distinguished by the high anti-carcinogenic and ant mutagenic potentials

Betacyanin components, betanidin and betanin overwhelms that in powder, which is consist of betanin as a dominating compound, instead of betanidin . An increase in the antioxidant capacity of blood serum in groups administered(Saber &

2006)

According to (Baz wane et al 2003) of all hydrocolloids in use today none has proven as a popular with the several public and found favor in a wide range of a food product as gelatin.

Gelatin is a proteinaceous materials obtained from animal connective tissue(collagen) (Steven, 2009). Gelatin is water soluble polymers that can be used as gelling, thickening or stabilizing agent; it is highly digestible protein, containing all amino acid except tryptophan and have low methionine, cysteine and tyrosine due to degradation during hydrolysis(Jamalia & Harvinder, 2002)

2.3.1. Effects of beetroot juice addition with yogurt

Studies investigating the addition of beetroot juice to yogurt have focused on its impact on sensory characteristics, nutritional composition, and bioactive content. Several research findings have suggested that beetroot juice addition can improve the color, flavor, and. Overall sensory acceptance of yogurt, providing consumers with a unique taste experience (Dhineshkkumar, 2016).

2.4. Types of Yoghurt

2.4.1 Set Yoghurt.

Traditional set yoghurt was made of concentrated milk, the milk was heated on an open fire until, and one third of water had evaporated. Then the milk was allowed to cool and when temperature of about 500c was reached the milk was inoculated with little

Abedimanesh, 2023).

Beetroot powder can be used as anti-cancer due to different nutrients containing mainly minerals, with polyphenals, flavonoids, dietary nitates and other useful nutrients help to prevent cancer and other related chemotherapy in addition perspectives actively support the renal system to overcome the adverse effects of Gentamicin (GM)'s primary and secondary reactive metabolites, resulting from the toxicantinduced damage (MEI Lan & Sahul Hamid, 2021).

Therefore, consumption of beetroot powder or other beetroot based beverages depicted positive implications by increasing the level of Superoxide Dismutase (as a primary antioxidant enzyme), and Catalase (involved in a detoxification procedure), while decreasing NO (with a controversial role in renal system), and oxidative stress, all as renal tissue-specific markers. Similarly, urea and creatinine content have lowered, while the protein profile of beetroot- based products accelerated, due to the action of bioactive compounds like betacyanins and betaxanthin(Wettasinghe, B, L, H, & Parkin, 2002).

CHAPTER THREE: METHODOLOGY\

3.1. Materials and equipment

- Gas cooker (Delonghi gas cooker made in china consume 4kw per day), Blender (browton blender made in china), Thermometer yoghurt similar process is still being used but either the milk is evaporated under vacuum or some milk juice is added, one may use some process for making set yoghurt from non-concentrated milk, the yoghurt obtained is less rich in flavor, is for less firm and is prone to syneresis and to enhance firmness' generally gelling agent is added to prevent syneresis and to enhance firmness especially if piece of fruit is added. This set should be made by homogenizing standardized milk to 550c 20mpa, pasteurization 850 c in 5min, cooling to 450c, inoculation with starter culture, 2.5%, packaging, incubate 2.5h then cool to 60c.

2.4.2. Firmness of Set Yoghurt

The firmness of yoghurt may vary based on the temperature where Firmness of yoghurt ranges from 90 to 10-4 g force for vat treatments, 74 to 90 g for high temperature furthermore short time treatments and 47 to 65 g for ultra-high temperature treatments

Firmness of set yogurt is often estimated by lowering a probe of given weight and dimensions into the product during a given time Its value depends on the method of measurement, especially the time scale, and on several product and process variables. Firms also depends on fat content, the higher fat content, the weaker gel because the fat globules interrupt the network, homogenizing of the milk, heat treatment of milk, yoghurt culture, incubation temperature, those are factor affect the firmness of yoghurt (Estelle, Parnell-Clunies, & Kakuda, 1986).

(thermometer, ranging in -49°C to 299.9°C made in china), Electrical weighing scale (originated from Zhejiang, China and Brand name is JINNUO, Model number: JF100C) SterillizedGlass jars/bottles (250ml and 500ml), Working clean table (made from stainless steel from china) Sauce pan, Oven (originated from china , Model XY-JZL-300: size 30×30×35cm) buckets, stirrer, dropper, sieves, Mixing bowls, Cutting board, Knives, Grinder , Rape PPE, spice bag, desiccator c, dishes, tongs, heat resistance gloves Stirrer, Dropper, Sieves, sizes, metal, Mixing bowls plastic, Cutting board , white, Knives sharp, PPE, spice bag, Desiccator cooling to room temperature , dishes .tongs, white Stainless steel table Working clean table (made from stainless steel from china, Sterilized Glass jars with white color, graduated numbers and capacity 2 l.

- Thermometer (thermometer, ranging in -49°C to 299.9°C made in china).
- Electrical weighing scale (originated from Zhejiang, China and Brand name is JINNUO, Model number: JF100C)
- bottles (250ml and 500ml), Sauce pan) buckets.

3.2. Experimental design and procedures

3.2.1. Preparation of raw materials

Red beetroot and tomatoes were selected based on maturity, color free from defects, free from any physical defects, flesh and firm with no signs of spoilage or decay, selected raw materials were cleaned under running water to remove any dirt or debris, then after Peeling and cutting by using sing a knife, peel the outer skin of red beetroot to remove any dirt, impurities and unwanted part. Washed red

2.4.3 Stirred Yoghurt

Stirred yoghurt, virtually always made from non-concentrated milk, after gel is formed, it is gently stirred to obtain a smooth and fairly thick, but still pourable, but they are other different in manufacturing process as set yoghurt is fermented after being packaged implying that final cooling has to be achieved in the package. Stirred yoghurt is almost fully fermented before it is packed and other different is that only certain strains of yoghurt produce the consistency or thickness after stirring and only so when incubation at a fairly low temperature. However, the bacteria make less of the desired flavor compounds at low temperatures. Stirred yoghurt process; standardization, homogenisationb550c 20mpa, pasteurization 5min 850c, cool to 30-320c, inoculation, incubation 16-20h, stirring, cooling 60c and packaging (Mancha & Ruiz-Gutierrez, 2021).

CHAPTER THREE: RESEARCH METHODOLOGY

3.1 Raw materials

Raw material and ingredients used were raw milk bought from Muhoza MCC, beetroot used bought from KARIYELI market, sugar, starter culture and stabilizers were taken from RP-IPRC Musanze food processing ingredients store room.

3.2. Tools and equipment

Balance: OEL display, Capacity: 135 g. Pan size: 95 mm, Response time: 5 seconds, beetroot was sliced into thin slices or small chunks and blend them using blender to facilitate the drying process. For tomatoes to remove outer skin was boiled in sauce pan at temperature of 50°C during 3 minutes and then after being boiled tomatoes were cooled and removal of outer skin followed e blending in blender and then cover them with aluminum foil in container.

3.2.2. Drying of beetroot

Red beetroot were dehydrated using oven set at 50°C Blended red beetroot were put in oven. 500 g of blended beetroot were dehydrating in the oven for 4 to 6 hours until they are thoroughly dry and crisp. Dried red beetroot were grinded into powder once and transfer them to blender and blend them into fine powder which sieved to obtain beetroot powder of good quality and then package obtained powder within airtight container waiting for further use in tomato ketchup and placed in cool, dry place away from direct sunlight to preserve the quality of the powder.

Obtained beetroot powder was used in making ketchup where ketchup formulation was shown in the table below The process of manufacturing tomato ketchup incorporated with red beetroot was done by using 1L of tomato puree with 0 ml of red beetroot powder which is named sample 0 and it was noted as control, sample 1: was 1 L of tomato puree with 50 g of red beetroot powder, sample 2: was 1L of tomato puree with 100 g of red beetroot powder

Table 3: proportion of ingredients

Sample name

Tomato puree

Weighty: 1kg, Minimum display: 0.01 mg, Measuring cylinder: plastic containers by Type: plastic, Capacity: 500 ml, Color: white, Graduated jags, they were characterized by Type: plastic, Capacity: 2 liters, Hot plate it could Cheat 1000C to 3000C, Stainless steel table It was also specified by Type: stainless steel, Size: 1x0.5 , refrigerator this was equipment that could cool 40C to 7 0 C which was also its specification, PPE: they were white in color Thermometer: It also has some specification like: Temperature range (-199 oC to1990C, Resolution: 0.01 0 C, Accuracy: +- 0.05, Battery: 3x1.5,) Lactometer: This was instrument specified with Length: 330 mm, Temperature standard; 60 0F, Temperature range: 20 0F to 100 0F, Grass graduated, pH meter was characterized by Compensation temperature: 0 to 1000 C and pH range of 1-14,visco-meter: was determined flow capacity or stickiness of yogurt, Wooden spoon: it was made in wood and it characterized by Type: Wood. Length: 150mm Sauce pan: it was characterized by Type: stainless steel and it, Filters: they were made with plastic covering layer ring, blender was specified by ability to blend 500 g, and electronic, Knives were stain less steel and sharp, cutting board was white and capable of used while cutting.

3.2. Research design

The aim of this study to produce yogurt by using beetroots juice as natural color, flavor ,acceptable viscosity and improve nutritional especially polyphenol. This research was carry

	in liters	Red beetroot powder (grams)	Vinegar	honey	Garlic7 onions
Sample 0	3L	0g	12 ml	30 ml	3 pieces
Sample 1	3L	50 g	12 ml	30 ml	3 pieces
Sample 2	3L	100g	12 ml	30 ml	3 pieces

3.3.3. Processing of tomato ketchup with beetroot powder

Ketchup were made by using ratios of ingredients according to each sample where control made without beetroot powder and processes followed in out four trials and one control

3.2.1 Treatment and amount of ingredient used.

In this research project five treatment SA with non -sweetened yogurt were used where treatment zero (T0) was control other was the sample added with juice in different amount of beetroots juice and water as shown in table below

Table 3: sample design un-sweetened yoghurt samples

Sample No	Milk quantity(ml)	Beetroots juice quantity(ml)	Water quantity(ml)	Cultures(g)	Stabilizer(g)
Sample A0	500	0	80	0.0116	4.35
Sample A1	500	10	70	0.0116	4.35
Sample A2	500	20	60		

this way cooking of tomatoes in a large pot after 3-4 min where seasonings were used are honey, salt, and vinegar and spice bag which is included garlic, celery and onions. By using gas cooker at the temperature of 45°C we added red beetroot powder and we continue cooking process over medium heat and bring the mixture to a gentle simmer. Once mixture is simmering, we reduced the heat to low (30°C) to maintain the simmer also stir the mixture occasionally to prevent sticking on the pot. Also cooking time was varied depending on the quantity of ingredients and the desired thickness. Then after we allowed ketchup to cool to room temperature before packaging into, bottles, once tomato ketchup incorporated with red beetroot powder was cooled and packed we transfer them into refrigerator.

3.4. Determination of Ash content

The amount of ash content was obtained by using method of association of official analytical chemists AOAC, where ketchup samples were weighed into porcelain crucibles which have been washed, dried in an air oven at 100 0 C for 30 minutes , cooled in a desiccator and weighed.

Samples of 2 g were placed inside a muffle furnace and heated at 500 oC for 5 hours, Samples were removed from muffle furnace and cooled in a desiccator and then weighed

$$\%\ Ash = \frac{A - B}{C} x \frac{100}{1}$$

A was weight of crucible+ ash, B was weight of crucible and C was weight of original sample(AOAC, 2010)

3.5. Determination of moisture content

				0.0116	4.35
Sample A3	500	40	40	0.0116	4.35
Sample A4	500	80	0	0.0116	4.35

five treatment SB with sweetened yogurt were used where treatment zero (T0) was control other was the sample added with juice in different amount of beetroots juice and water as shown in table below

Table 4: sample design for sweetened yoghurt

Sample N0	Milk quantity (ml)	Beetroots juice quantity(ml)	Water quantity(ml)	Sugar(g)	Stabilizer(g)	Culture(g)
Sample B0	500	0	80			

The moisture content was determined by using the method of AOAC where porcelain crucibles were washed and dried in the air oven at 1000 C for 30 minutes and allowed to cool within desiccator. Two grams of tomato ketchup incorporated with beetroot powder were placed into weighed crucibles and then put inside the oven set at 105 0 C for 4 hours The sample were removed from the oven after that period and cooled where they were weighed this means both solids and liquid samples. The drying was continued and all samples with the crucibles were weighed until a constant weight was obtained. Therefore the % moisture content was calculated as shown in the equation below

$$\% \text{ Moisture content} = \frac{A - B}{A} X \frac{100}{1}$$

Here A was considered as weight of original sample and B was considered as weight of dried sample

3.6. Sensory evaluation

To ensure the sensory properties of ketchup samples had been changed or not, sensory analysis was taken into consideration where the number panelist used were 9 from RP-IPRC Musanze students in food processing and some staffs, and that panels judged and assess the sensory quality of tomato ketchup incorporated with red beetroot powder. In addition sensory analysis of tomato ketchup with red beetroot powder was done by determining the following sensory attributes such as aroma, texture, colour, taste and overall acceptance and the test used was 9 point hedonic scale, which varied in range from extremely like score 9 to extremely dislike score

40.6
4.35
0.0116
Sample B1
500
10
70
40.6
4.35
0.0116
Sample B2
500
20
60
40.6
4.35
0.0116
Sample B3
500
40
40
40.6
4.35
0.0116
Sample B4
500
80
0
40.6
4.35
0.0116

3.3. Production Beetroots juice

Beetroot juice was firstly made where beetroots were selection beetroots and cleaned in clean

CHAPTER FOUR: RESULTS AND DATA INTERPRENTATION

4.1. Data Analysis Method

4.1.1. Data Collection

Data was collected by using questionnaire where the selected randomly panelist and gave marks according to the sensory attributes such as colour, flavour, texture, taste and overall acceptability of the samples that were given to them.

4.1.2 Data Interpretation

After testing the acceptance of the samples from the panelist data collected were analyzed by micro soft Excel vision 11 using Analysis of Variance (ANOVA) method in order to test the significant difference between the samples at the level of $p<0.005$

4.2. Sensory evaluation results description

During conducting this research sensory evaluation was one of things taken into consideration where the obtained data showed significant difference among the products where sample 1 was the most liked sample which was ketchup contained beetroot powder as shown in the figure 1 below

Figure 1: Sensory evaluation data

water into 3 times, after washing were measure 300g of washed beetroots and 300 clear water cut and blended to obtained beetroot juice which underwent filtration after obtaining beetroot

pulp after filtration was determined pH and brix content by using refractometer and pH meter

Figure2: production of beetroot juice

3.3.1. Processing of yoghurt samples

In making yoghurt samples there was two categories sweetened yoghurt samples and nonsweetened yoghurt samples. For non-sweetened yoghurt samples considered as samples SA1 there control was first sample made where raw milk received and tested for checking its quality like pH, density and its organoleptic, then filtered after measure 500 ml for each sample and 80 ml of water, were heated to 90 0 C but while pasteurization at 700 C stabilizers were measured and added at that temperature then mixture heated to 90 0C, milk cooled to temperature of 450 C where starter culture were added and milk were put in stainless steel containers and sent into incubator which was set at 450C for 6 hours after that period yoghurt obtained mixed to break some fat globules and then packaged and labeled as sample SA0 for A , finally it was stored in in refrigerator at 4-7 0C.

For all yoghurt sample in SA (non- sweetened yoghurt samples) SA1, SA2, SA3. SA4 the

The results obtained showed that sample 1 was liked by panelists where the sample 2 was second sample liked and sample 0 which was control was third sample liked by panelist. Table 6 showed that among those samples there was significant difference among ketchup samples where p value obtained was 0.010829

4.3. Laboratory results on Ash content and moisture content

4.3.1. Ash content

Obtained results of content were shown in the table 4 and the figure 2 following was made to show clearly how the Ash content varied as the amount of beetroot powder increased

Table 4Results of Ash content

Samples	Ash content
Sample 0	2.7
Sample 1	4.9
Sample 2	6

The figure below showed how Ash content varied among the samples

Figure 2: results of Ash content

4.3.2. Moisture content results

In manufacturing ketchup sample there were samples contained beetroot powder and the results obtained showed that as the amount of beetroot powder decreased the moisture content of ketchup procedures used to make them were the same but the difference was the amount of beetroot juice used and amount of water used as shown in the Table 3 but other things were the same. For samples SB the ways used to make samples were the same as SA except during pasteurization where on control sample for A which did not contain sugar, but other samples while pasteurizing sugar were added according to the ratios shown in table 4 and was added at temperature of 70 0 C then steps followed as used in section A samples and labeled accordingly SB1, SB2, SB3 and SB4'

steps milk reception

filtration

mixing milk with beetroot juice

pasteurization at 700c and addition stabilizer

pasteurization at 900cat 15 min

cooling at 450c

inoculation at 450c

Incubation at 450c at 6hour

Homogenization

Packaging in bottles

samples,, That means that when the amount of beetroot powder increased the moisture content increased.

Table 5: Obtained moisture content

Samples	Moisture content	Amount of red beetroot powder
Sample 0	81%	0 %
Sample 1	7.5%	50 %
Sample 2	69%	100 %

The figure was made to show how the moisture content level varied based on how amount of beetroot powder used increased

Figure 3: Results of Moisture content CHAPTER FIVE: CONCLUSION AND RECOMMENDATION

5.1. Conclusion

This research was conducted with aim of making tomato ketchup incorporated with red beetroot powder, with objective of improving nutrition content and sensory attributes and research reached on the objective because the proposed ketchup samples obtained and liked by panelists where significant different among the product was p value of 0.010829 which was in the range of acceptable p value of less than p value of 0.05. In addition the laboratory analysis showed the

Storage and cooling at 40c

3.4. Determination of viscosity in yogurt added with beetroot juice

The test tube and spindle used were cleaned and using cleaned water, detergent and brushes before used at next sample and The book field viscometer was assembled and calibrated in order to reduce and avoid bias or errors in its working as setting time of 1 minute, rpm 100, and spindle 21, beetroot Yoghurt samples were taken and their temperature was measured temperature to make sure if it reaches with 20.3 0 C, Prepared yoghurt sample was put into non-graduated test tube and also was put into each place before press the start bottom of this instrument to start working, The values of viscosity obtained were recorded into cp unit, Torque present into % unit. The records were recorded three times in order to make mean calculation easily with used three numbers value near for each other's or use three at all, Calculations of viscosity leading value and torque called Mean were made, Viscometer was disassembled and cleaned where all parts required to be cleaned were cleaned before its safe storage and put in its packaging material

3.5. To determine the presence polyphenol in yogurt added with beet root juice

The total phenolic compounds were determined using the rapid test. The phenolic compounds was oxidized to phenolates by the reagent at

difference also among the product where Ash content determination test showed that sample 2 had higher ash content value which was 6,sample 1 was the second sample with ash content of 4.9 and sample 0 which was considered as control had ash content of 2.7, addition of red beetroot powder also affected the moisture content level where moisture content decreased as beetroot powder increased in ketchup samples sample 0, sample 1 and sample 2 with moisture content of 81%, 75% and 69 respectively. Addition of beetroot powder in tomato ketchup manufacturing is possible and workable

5.2. Recommendation

Based on the obtained results in our research we recommend the following:

Red beetroot powder truly can be used in manufacturing tomato ketchup as an ingredient

We recommend researchers to continue the research on how red beetroot powder can be used in other food products. We recommend also researchers on to make research on the nutritional profile of tomato ketchup incorporated with red beetroot powder mainly fiber content, folate and vitamin C

We recommend RP-IPRC Musanze to provide support to students in the running their final year project by giving them required materials and help them to conduct laboratory analysis or equipping campus food laboratories with all required reagents, instruments, tools and equipment.

5.3. Limitation

alkaline pH in a saturated solution of sodium carbonate resulting in a blue molybdenum-tungsten complex in addition the Folin ciocalteu reagent, was diluted 10 times (2.5 ml) and 2 ml of saturated sodium carbonate (75 g/l) and 50 µl of sample (diluted ten times) was mixed and homogenized for 10 s and heated for 30 min at 45ºc. The absorbance at was taken into consideration where the read after cooling to room temperature. The absorbances of each sample SB0, SB1, SB2, SB3 and SB4.

3.6. Sensory evaluation of yogurt added with beet root juice

The sensory evaluation conducted by using affective test to ensure the preference and acceptance of beetroot yoghurt samples where the following characteristics were taken into consideration:

- Appearance
- Aroma
- Flavor
- Texture, and
- Overall acceptability

During our sensory analysis; the panelists used were trained, where 5 point's hedonic scale at duplicate was taken into consideration. A panel of judges assessed sensory characteristics of control yoghurt, and of beetroot yoghurt on each and every treatment. Sensory evaluation carried out to determine the organoleptic

This project was limited to laboratory analysis where the folate, vitamin C and fiber content did not be analyzed due to the problem of lacking money, and some equipments and instruments which are not working properly in addition in campus laboratory there was no required reagents should be used for running laboratory analysis for analyzing folate , vitamin C and fiber content

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characteristics in terms of color, aroma, taste, texture, and overall acceptance of all treatments. In addition A Five-point hedonic scale which varied from like very much score 5, to dislike very much 1 were taken into consideration. A total of trained 5 panelists from RP IPRC Musanze including student and staff participated in the study

To avoid bias or any errors may occur; Samples were also coded with 3 digits so that the panelist could identify them. In addition the sample presentation order was randomized among and within assessors. The selection of panelists was chosen based on the availability and motivation

CHAPTER FOUR: RESULTS AND INTERPRENTATION

4.1. Results of juice extraction

In beetroot juice extraction resulted to 300 g from of beetroot mixed 300 ml of water where its brix content was 5.3 0 Brix and pH was 5.9 then after obtained juice proceeded to the further use in yoghurt which was 450 ml of juice divided into parts 150 for sweetened yoghurts and 150 for non-sweetened yoghurt.

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4.2. Effect of adding beetroot juice on physicochemical properties of yoghurt

4.2.1. Viscosity

Addition of beetroot juice in the yoghurt affected viscosity where viscosity obtained was varied according to the amount of beetroot juice used and resulted variation in sweetened and non-sweetened samples

Non sugar yoghurt sample their viscosity based on the amount of beetroot pulp used

Figure 2: Results of viscosity in SA1

Obtained results showed that Sample1 and Sample4 had the higher viscosity than other samples where an increase of beetroot juice should increase or decrease viscosities depend on the ratio used. But addition of small amount of beetroot juice increase the viscosity of yoghurt

Figure 3: Results of Viscosity for SB

The results showed that the obtained the value of viscosity changed in different ways due to the amount of beetroot juice used. The addition of small amount of juice affect viscosity where increased.

4.3 Effect of beetroot juice on polyphenols of yoghurt

The polyphenols obtained are showed in the table below where each sample contained polyphenols based on the amount of beetroot

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APPENDIX:

Appendix 1: Form used during Sensory Evaluation juice used.

Table 5: results of phenolic compound

Samples	Polyphenols
Sample A 0	Absent
Sample A2	Absent

The method used is rapid teste

4.4. Effect of beetroot juice on sensory characteristics of yoghurt

A. Color

The results obtained from sensory evaluations showed that panelists like sample 3 based on their color where the 5 was the highest score obtained in color which was belong to sample 2 where on the samples of the liked sample also was sample 2 which had the score of 5 among those samples the liked sample was sample 2 in non-sweetened as well as sweetened samples.

B. Smell

The results obtained showed that in sweetened samples sample 2, sample 3 and sample 1 were the most liked samples with highest score of 5 and addition of beetroot juice in yoghurt making increased smell of yoghurt samples. For no sweetened yoghurt samples, sample 2 was the most liked sample which had score of 4.8 in general in all samples sample 2 in sweetened yoghurt was the most liked sample

C. Taste

In sweetened sample the obtained results

Code of the Panelist: ……………………………………….

Sample: …………………………………………………..

Sensory attributes of tomato ketchup with beetroot powder are the major elements which determine the acceptability of it by consumers. Sensory analysis is actually considered as a requirement before a product goes out of the industry to the market.

Instructions

1. Fill this form correctly and follow instructions
2. Please rinse your mouth after one sample tasting with potable water
3. Do not seek another person's option
4. Observe and taste these doughnut samples and rate them in the provided boxes how much you like or dislike them.
5. Use the appropriate scale to express your feeling by writing the number corresponding to your feeling on the sample attributes in the table on the right side of the sheet.
6. Your feeling may be like or dislike, feel free and be honest to express your personal feeling.

This is a study and it is confidential.

Appendix 2

Table 4: sensory evaluation table

Characteristics

showed that the most likes sample was sample 2, sample 3and control which had score of 5 in taste that was different to no sweetened yoghurt sample where the most liked sample 3 was with 3.4 score, that means that in general sample 2, 3, and sample 0 in sweetened yoghurt samples were the most liked samples in taste attribute.

D. Texture and mouth feel

The results obtained from sensory evaluation showed that in non-sweetened yoghurt samples the obtained scores showed that sample 2 and sample 1 had the highest score which was 4.8 but was in sweetened yoghurt samples texture and mouth feel results showed that sample 2, and control had the most liked score because they had 5 which was the highest score.

E. Overall acceptability

The results obtained showed that in non-sweetened sample 2 and sample 1 were the most liked samples where they had highest score of 4.8 in addition sweetened yoghurt samples sample 2 was the most liked sample where it had high score of 5, generally in all samples sample 2 in sweetened yoghurt was the most liked sample with score 5.

Figure 4: Results of sensory evaluation for SA1 samples

The results of sensory evaluation showed that

appearance
Aroma
Taste
Texture and mouth feel
Overall acceptability
Dislike extremely(1)
Dislike very much (2)
Dislike moderately (3)
Dislike slightly (4)
Neither like nor Dislike (5)

SA1 was the most liked sample in nonsweetened sample. Because the obtained data in samples which did not contain sugar that there was significant different because p value obtained was 0.005477 which was below acceptable p value < 0.05.

4.3. Sensory evaluation for sweetened yoghurt

The data below are data of sensory evaluation for sweetened sample

Figure 5: Sensory evaluation for sweetened yoghurt for SB2

Obtained results showed that SB2 was the most liked sample among all attributes where it had highest score in color, smell, taste and texture where SB3 which had highest score in smell and taste

The obtained data showed that there was significant difference among products where p value obtained was below 0.05 which was 0.003429

CHAPTER FIVE: CONCLUSION AND RECOMMENDATION

5.1. Conclusion

Frankly manufacturing yoghurt using beetroot juice either yoghurt can be considered as plain

Like slightly (6)

Like moderately (7)

Like very much (8)

Like extremely (9)

Appendix 2: Image of buying beetroot from the market or sweetened yoghurt is applicable and possible where manufacture yoghurt samples among all samples liked by panelists but in different ways, for SA1 samples sensory analysis results obtained showed that there was significant different because panelists like them on the same close levels where p Value obtained was 0.005477 which was below acceptable p value for significant different. Furthermore for nutritional analysis results showed that samples for S A1 varied in the viscosity where S0 had 0.925, S1 had 1.199, S2 had 0.973 S3 had 1.061 and S4 had 1.077 of viscosity for SB2 samples the results of sensory evaluation showed that there was significant different also where obtained value was p value of 0.003429 which was in the acceptable p value <0.05 for laboratory analysis the results obtained showed that products also varied in the viscosity whereS0 had 1.9955, S1 had 2.106, S2 had 1.718, S3 had 1.304 and S4 had 1. 6525 of viscosity. For analysis of polyphenols in yogurt added with beetroot juices SA0 had absent and SA2 had absent.

5.2. Recommendation

According to the results obtained in this study we recommend the following:

Beetroot juice should be used in processing of yoghurt as coloring and flavoring agent. We recommend RP/IPRC Musanze to support and provide all facilities to students in order to improve the research culture and recommend Agriculture engineering department mainly

Appendix 3: Ash content analysis preparation

Appendix 4: Tomato ketchup incorporated with beetroot powder preparation

Table 6: Sensory evaluation results of sample 0 and sample 1

Sample 0 (S0), Code 789

Sample 1 (S1) code 456

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program of Food Processing to continue to	8	
improve the existing recipes with innovation in	8	
order to make many uses of animal and	9	
agriculture yields for benefiting human health.	8	
We recommend the researchers to continue	8	
their findings on how to utilize beetroot juice in	9	
production of other food or dairy products. We	9	
recommend also the researcher to conduct	9	
nutritional analysis of yoghurt enriched with	9	
beetroot juice mainly Vitamin C, and Fibers.	9	
Limitation	3	
Based on the method of analysis the laboratory	7	
used was not available in our college, they are	8	
affected the amount of money carry out in	8	
laboratory test another company. Due to low	6	
capacity were not carry out all samples in	7	
laboratory test the tests was done is control	9	
and sample 3 that are non –sweetened yogurt.	9	
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Reference	7	
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development of yoghurt Flavoured with	9	
Beetyroot juice (Beta vulgaris L). International	9	
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Jamalia, & Harvinder. (2002). properties of	8	
gelatin from skins of fish black tilapia and red	9	
tilapia. journal of food chemistry, 81- 84.	9	
kavitha, R., & Smetanska, I. (2020). Chapter	9	
19- Red beet. In Nutitional composition and	9	
antioxidant properties of fruits and vegetables	9	
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alcholic extract and its main constituent,	9	
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the potential of red beetroot supplementation in	A	
health and disease. Nutrients, 7(4), 2801-2822.	N	
	7.2	
Verbeken, e. A. (2006). Interaction between K-	8.2	
Carrageenan, Milk protein and modified starch.	8.4	
International dairy journal, 482-488.	6.2	
Wettasinghe, M., B, B., L, P., H, X., & Parkin,	7.2	
K. (2002). Phase II enzyme-inducing and	9	
antioxidant activities of beetroot (Beta Vulgaris	9	
L) extracts from phenoltype of different	9	
pigments. Journal of Agriculture and food	9	
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Whittaker, G. (2023). Beetroot benefits for men:		

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Wlodarczyk, J., & Janiszewska, E. (2013). influence of spray drying conditions on beetrooty pigments retention after microencapsulation (2 ed., Vol. 20). landon: Acta- agrophysica.

Appendix:

PRODUCT

Table 6: sensory evaluation results for S0 and S1 in non-sweetened yoghurt

Sample 3 (S3), Code 123

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Sample 0 (S0), Code 589	7
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Sample 1 (S1) code 153	8
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4	8
5	8
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2	7.5

2 4 3 5 4 4 5 3 5 5 3 3 5 4 5 4 4 5 4 5 5 4 2 5 3 5 5	7.2 6.2 7.3 Table 7: Sensory evaluation results of sample 3 Table 8:ANOVA table for sensory evaluation Anova: Single Factor SUMMARY Groups Count Sum Average Variance

4 4 1 4 4 5 2 5 3 5 4 4 5 2 5 4 2.8 4.6 3 4.6 3.6 4 4.6 2.8 4.8	Sample 0 5 37.2 7.44 0.788 Sample 1 5 45 9 0 Sample 2 5 37.2 7.44 1.013 ANOVA

4.6

Table 7: sensory evaluation results for S3 and S4in unsweetened yoghurt samples

Sample 2 (S2), Code 103

Sample 3 (S3) code 214

Source of Variation	SS	Df	MS	F	P-value	F crit
Between Groups	8.112	2	4.056	6.756247	0.010829	3.885294
Within Groups	7.204	12	0.600333			

1 5 4 3 4 4 4 5 4 5 4 2 5 5 3 5 5 4 5 4 4 5 3 5 5	Total 15.316 14 ii 11

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Table 8: results of sensory evalutaion for S4 in unsweetened samples

Sample 4 (S4), Code 593

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3.6

Table 9: Sensory evaluation results for S0 and S1 in sweetened samples

Sample 0 (S0), Code 871

Sample 1 (S1) 654

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4 5 5 4 4 3.8 4.8 5 5 4.8 4.2 5 4.8 4.6 4.4 Table 10: Sensory evaluation results for S2 and S3 in sweetened yoghurt sample Sample 2 (S2), Code 482 Sample 3 (S3) 694	

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5 5 5 5 5 5 5 5 5 4 5 4.8 4.8 4.8 4.8 4.8 4.6 5 5 4.8 4.8 Table 11: sensory evaluation results for S4 in section B	

Sample 4 (S4), Code 121 1 3 4 5 5 4 2 4 4 4 4 4 3 3 4 5 5 3 4 4 3 5 5 5 5 3	

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3.8

Table 12: Anova table of sensory evaluation for section B

Anova: Single Factor

SUMMARY

Groups Count Sum Average Variance

Sample 0 5 19.2 3.84 0.328 Sample 1 5 23 4.6 0.1 sample 2 5 24 4.8 0

sample 2 5 24.2 4.84 0.028

sample 4 5 20.6 4.12 0.412

ANOVA

Source of Variation SS Df MS F P-value F crit

Between Groups 3.888 4 0.972 5.599078 0.003429 2.866081

Within Groups 3.472 20 0.1736

Total 7.36 24

Table 13: Anova table for sensory evaluations of Section A samples

Anova: Single Factor

SUMMARY

Groups

Count

Sum

Average

Variance

Sample 0

5

19.2

3.84

0.328

Sample 1

5

23

4.6

0.1

sample 2

5

24

4.8

0

sample 3 5 24 4.8 0.08 sample 4 5 20.6 4.12 0.412 ANOVA Source of Variation SS Df MS F P-value F crit Between	

Groups

3.7344
4
0.9336
5.073913
0.005477
2.866081
Within Groups
3.68
20
0.184

Total
7.4144
24

ii

11